

CityAQI Forecast

Machine Learning-Based Short-Term AQI Forecasting System

Internship Project Report

Developed as part of the

10Pearls Shine Internship Program - Cohort 9

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Learning Path: Data Sciences

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1. Introduction

Air pollution is a major environmental concern, making timely air-quality monitoring and forecasting important for public health. **CityAQI Forecast** is a machine learning-based system that predicts Air Quality Index (AQI) for 24, 48, and 72 hours across 12 cities in Pakistan.

Developed as part of the **10Pearls Shine Internship Program**, the project uses weather and air-quality data from the Open-Meteo API. The system performs automated data collection and feature engineering, stores features in the Hopsworks Feature Store, and trains separate Gradient Boosting models for each forecasting horizon. Live predictions are generated using current environmental data, with SHAP providing explanations for model predictions.

The final system is delivered through an interactive Streamlit dashboard and supported by GitHub Actions for automated feature updates and model retraining. The application enables users to view current conditions, future AQI forecasts, and prediction insights for supported cities.

1.1 Objectives

The main objectives were to:

- Collect and process weather and air-quality data automatically.
- Develop 24-hour, 48-hour, and 72-hour AQI forecasting models.
- Use Hopsworks for feature and model management.
- Implement live inference and SHAP-based explainability.
- Develop and deploy an interactive Streamlit dashboard.
- Automate data updates and model retraining through CI/CD.

2. Project Overview

2.1 Problem Statement

Current AQI monitoring provides information about existing air-quality conditions but does not indicate how pollution levels may change over the coming days. This limits the ability of users to anticipate deteriorating air quality and make timely decisions.

2.2 Proposed Solution

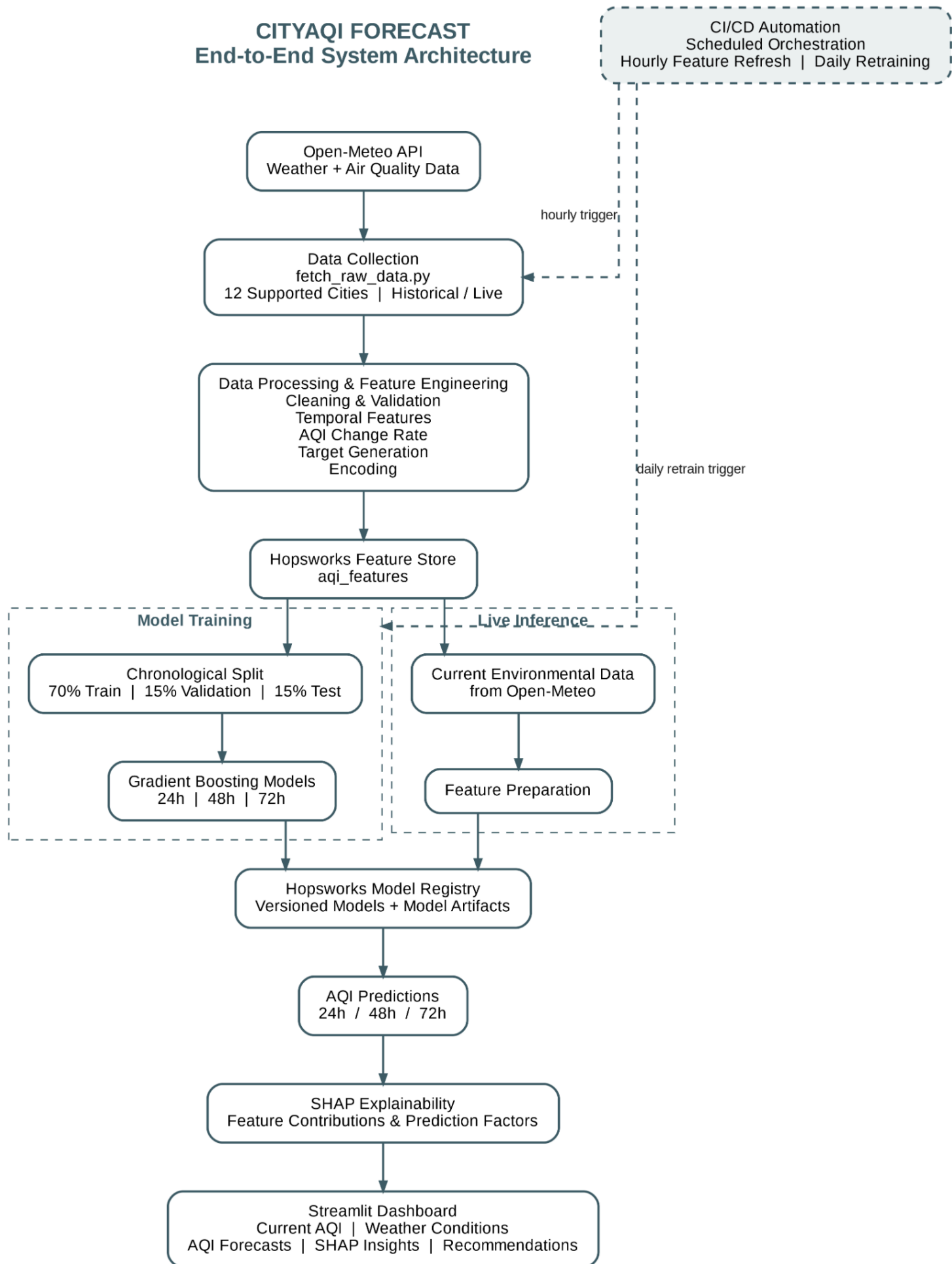
CityAQI Forecast addresses this gap by using historical and current environmental data to generate short-term AQI forecasts for **24, 48, and 72 hours**. The system combines automated data processing, machine learning, and a web-based interface to provide accessible air-quality forecasts for multiple cities.

2.3 Technology Stack

Technology	Purpose
Python 3.11.7	Application and ML development
Open-Meteo	Weather and air-quality data
Pandas / NumPy	Data processing
Scikit-learn	Machine learning
Hopsworks	Feature Store and Model Registry
SHAP	Model explainability
Streamlit / Plotly	Dashboard and visualization
GitHub Actions	CI/CD automation
Git / GitHub	Version control

3. System Architecture

CITYAQI FORECAST End-to-End System Architecture



4. Data Processing and Feature Engineering

4.1 Data Collection

The system uses the Open-Meteo API to collect hourly weather and air-quality data for 12 cities in Pakistan. Weather variables include temperature, humidity, atmospheric pressure, wind speed, precipitation, cloud cover, and visibility, while air-quality data includes AQI and major pollutants such as PM2.5, PM10, carbon monoxide, nitrogen dioxide, sulphur dioxide, and ozone.

4.2 Feature Engineering

The collected data is processed and transformed into features suitable for machine learning. Time-based features such as hour, day, month, and weekday are extracted from timestamps. Additional features, including AQI change rate, temperature change, and humidity change, are calculated to capture recent environmental trends.

Cyclical encoding is applied to hour and month to represent their periodic nature, while categorical variables such as city and weekday are one-hot encoded. Features that are not used by the final models are removed during preprocessing.

4.3 Forecast Target Generation

Separate target variables are generated for each forecasting horizon by shifting AQI values within each city:

- **Target 24h:** AQI 24 hours ahead
- **Target 48h:** AQI 48 hours ahead
- **Target 72h:** AQI 72 hours ahead

This produces three forecasting tasks that are trained and evaluated independently.

5. Machine Learning Model Development and Selection

5.1 Dataset Preparation

The final dataset contained **103,104 records** from 12 cities. After preprocessing, there were **36 processed columns**, consisting of **35 model input features** and one time column. The three forecast targets (24h, 48h, and 72h) were kept separately and were not used as input features.

The data was divided chronologically into 70% training, 15% validation, and 15% test sets, resulting in **72,122 training records**, **15,455 validation records**, and **15,455 test records**. The test set remained untouched during model selection.

5.2 Baseline Model Comparison

To explore multiple machine learning approaches, three regression models were evaluated independently for each forecasting horizon:

- Ridge Regression
- Random Forest
- Gradient Boosting

The validation results were:

Horizon	Model	RMSE	MAE	R ²
24h	Ridge Regression	18.8455	12.5010	0.4566
	Random Forest	17.9555	11.5724	0.5067
	Gradient Boosting	16.7810	10.5032	0.5692
48h	Ridge Regression	25.1470	16.6947	0.0004
	Random Forest	25.4787	18.1482	-0.0262
	Gradient Boosting	21.7833	15.0486	0.2499
72h	Ridge Regression	26.0705	17.6619	-0.1109
	Random Forest	28.0655	20.7937	-0.2874
	Gradient Boosting	23.1266	17.0017	0.1258

Gradient Boosting achieved the best RMSE, MAE, and R² for all three forecasting horizons. Based on this comparison, it was selected as the primary model family for further experimentation.

5.3 Hyperparameter Tuning

Gradient Boosting was further evaluated using 8 hyperparameter combinations with 3-fold cross-validation, varying `n_estimators` (100, 200), `learning_rate` (0.05, 0.1), and `max_depth` (2, 3).

For the 24-hour forecast, tuning produced a small improvement. The selected configuration was `learning_rate=0.1`, `max_depth=2`, and `n_estimators=200`, achieving **RMSE 16.6824**, **MAE 10.5118**, and **R² 0.5742** on the validation set.

For the 48-hour and 72-hour forecasts, tuning did not improve the overall validation performance, so the baseline Gradient Boosting model was retained.

Since AQI forecasting is time-dependent, **TimeSeriesSplit** was additionally tested for the 48-hour horizon. The tuned result still did not outperform the baseline, so no further tuning was performed.

5.4 Final Model Selection

The final configurations were therefore:

Horizon	Final Model	Configuration
24h	Tuned Gradient Boosting	<code>learning_rate=0.1</code> , <code>max_depth=2</code> , <code>n_estimators=200</code>
48h	Baseline Gradient Boosting	Default baseline
72h	Baseline Gradient Boosting	Default baseline

The reported values above are **validation-set results**, not final test results. The 15,455-row test set was kept untouched for final evaluation.

Overall, performance decreased as the forecasting horizon increased. Gradient Boosting remained the strongest approach among the evaluated models.

The selected models were subsequently registered in the Hopsworks Model Registry for live inference.

Other potentially suitable approaches, including **XGBoost** and **LSTM**, were also considered for experimentation. However, due to the available project timeline, these models were not implemented and evaluated. The model selection was therefore based on the three approaches that were practically evaluated: Ridge Regression, Random Forest, and Gradient Boosting.

6. Feature Store, Model Registry and Automation

6.1 Hopsworks Feature Store

The processed features were stored in **Hopsworks Feature Store** using a feature group named `aqi_features`. The Feature Store provides a centralized and structured location for storing the data used by the training and inference pipelines.

The feature pipeline supports both **historical backfilling** and **hourly updates**. During each update, newly collected environmental data is processed, features are generated, validated, and then ingested into the Feature Store.

6.2 Model Registry

The selected Gradient Boosting models were registered in the **Hopsworks Model Registry** as separate models for the three forecasting horizons:

- `AQI_GB_24h`
- `AQI_GB_48h`
- `AQI_GB_72h`

Each registered model includes its trained model and supporting artifacts required for inference. Model versioning allows updated models to be registered and managed without disrupting the existing inference workflow.

6.3 Automated Pipelines

GitHub Actions was used to automate the project's recurring workflows. Two scheduled workflows were implemented:

- **Hourly Feature Pipeline:** collects the latest weather and air-quality data, performs feature engineering, and updates the Hopsworks Feature Store.
- **Daily Training Pipeline:** retrieves updated historical features, retrains the models, evaluates them, and registers new model versions when appropriate.

This automation reduces manual intervention and keeps the feature data and models updated over time.

7. Live Inference and Explainability

7.1 Live Inference

The system retrieves current weather and air-quality data from the Open-Meteo API for the selected city and prepares it using the same feature-processing pipeline used during training.

The inference pipeline loads the latest models from the Hopsworks Model Registry and generates AQI forecasts for **24, 48, and 72 hours**. It also verifies feature compatibility before prediction, after which the results and current conditions are passed to the Streamlit dashboard.

7.2 SHAP-Based Explainability

To make the model predictions more interpretable, **SHAP (SHapley Additive exPlanations)** was integrated into the inference pipeline. SHAP calculates the contribution of individual features to each prediction and identifies the features having the greatest influence.

The system presents the **top five contributing features** for a prediction and indicates whether each feature contributed toward an **increase or decrease in the predicted AQI**.

This provides users with more context about the forecast instead of presenting the AQI prediction as a standalone value.

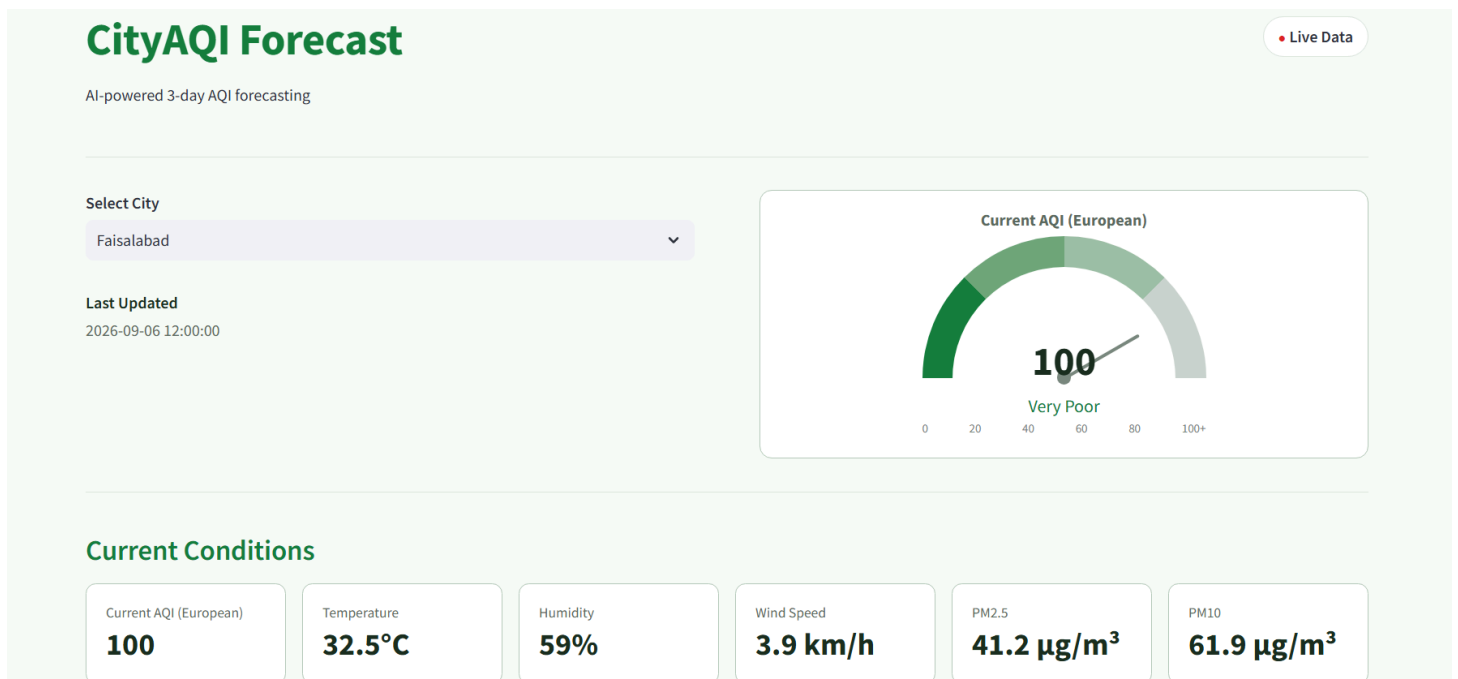
8. Dashboard, Deployment and Results

8.1 Streamlit Dashboard

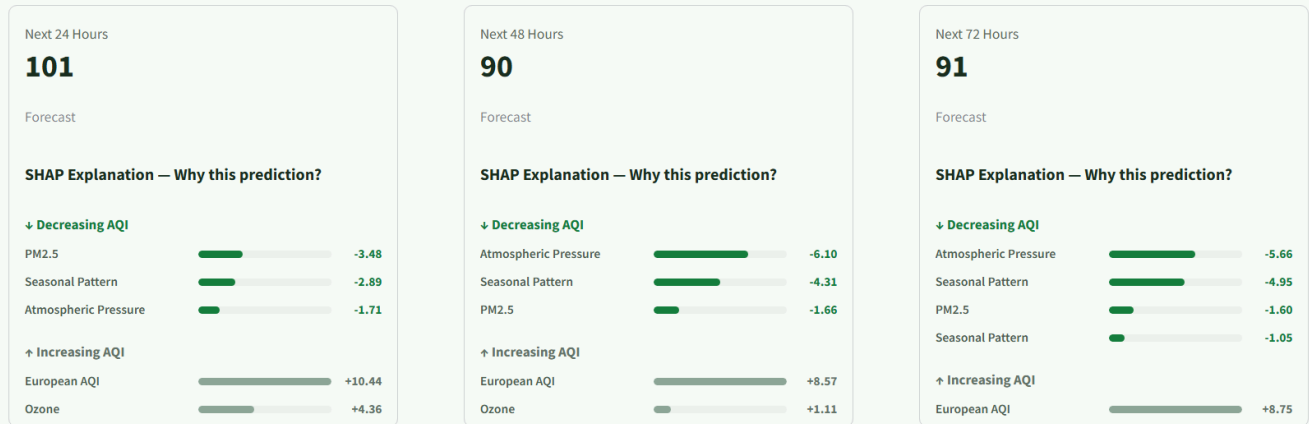
The system was developed as an interactive **Streamlit dashboard** where users can select a supported city and view its current environmental conditions and AQI forecast.

The dashboard provides:

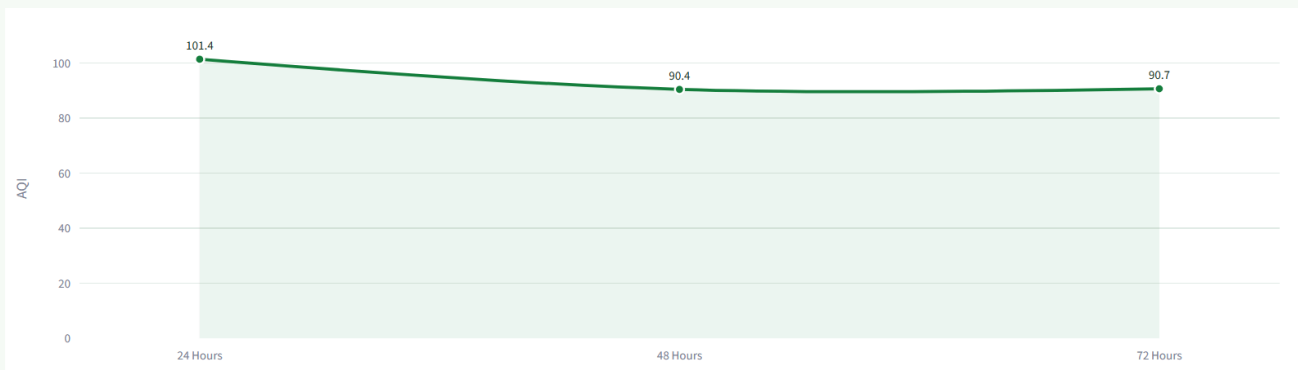
- Current AQI, temperature, humidity, wind speed, PM2.5, and PM10.
- 24-hour, 48-hour, and 72-hour AQI forecasts.
- SHAP-based prediction insights showing the main contributing factors.
- Forecast visualization using interactive charts.
- AQI category and corresponding recommendations.



3-Day AQI Forecast with SHAP Explainability



AQI Forecast



Air Quality Status

AQI Category Very Poor	Recommendation Health effects may occur. Sensitive individuals should avoid prolonged outdoor activity.
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AQI Predictor • AI-powered 3-day air quality forecasting

8.2 Deployment

The completed application was deployed using **Streamlit**, making the dashboard accessible through a web browser without requiring local setup. The source code, project structure, documentation, and deployment configuration are maintained in GitHub.

The project also uses **GitHub Actions** for automated feature updates and daily model retraining, providing a complete workflow from data collection to model deployment and user-facing predictions.

9. Challenges, Limitations and Future Work

9.1 Challenges

Several technical challenges were encountered during development, including handling API reliability and data consistency, integrating the Hopsworks Feature Store, and maintaining compatibility between training and live inference features. Model performance also became more challenging as the forecasting horizon increased.

9.2 Limitations

The current system is limited to 12 Pakistani cities and depends on the availability and quality of data from the Open-Meteo API. The 72-hour forecast has limited predictive strength, indicating that longer-term AQI prediction remains challenging with the current features and models.

9.3 Future Work

Future improvements could include:

- Exploring **XGBoost** and **LSTM** models.
- Using larger and longer historical datasets.
- Expanding coverage to more cities and regions.
- Incorporating additional environmental and meteorological features.
- Improving long-horizon forecasting performance.

10. Conclusion

CityAQI Forecast successfully integrates automated data collection, feature engineering, machine learning, feature and model management, live inference, explainability, and deployment into a complete AQI forecasting system. The evaluation of multiple machine learning models led to the selection of Gradient Boosting for 24-hour, 48-hour, and 72-hour forecasting, while SHAP provides interpretable prediction insights. With Hopsworks, GitHub Actions, and Streamlit integrated into the workflow, the project demonstrates an end-to-end approach to building and deploying a practical, automated machine learning application for short-term AQI forecasting.

Project Links

- GitHub Repository: <https://github.com/Areej-Shoaib/10pearls-internship-project>
- Live Streamlit Application: [CityAQI Forecast · Streamlit](#)